

TECHNICAL REPORT

Automated Conveyor Belt System with Sensor Integration

1. System Overview

The automated conveyor belt system is designed to transport, detect, count, sort, and safely deliver products along a manufacturing or logistics line without manual intervention. A PLC serves as the central processing unit, continuously polling sensor inputs and issuing commands to output devices based on programmed ladder logic.

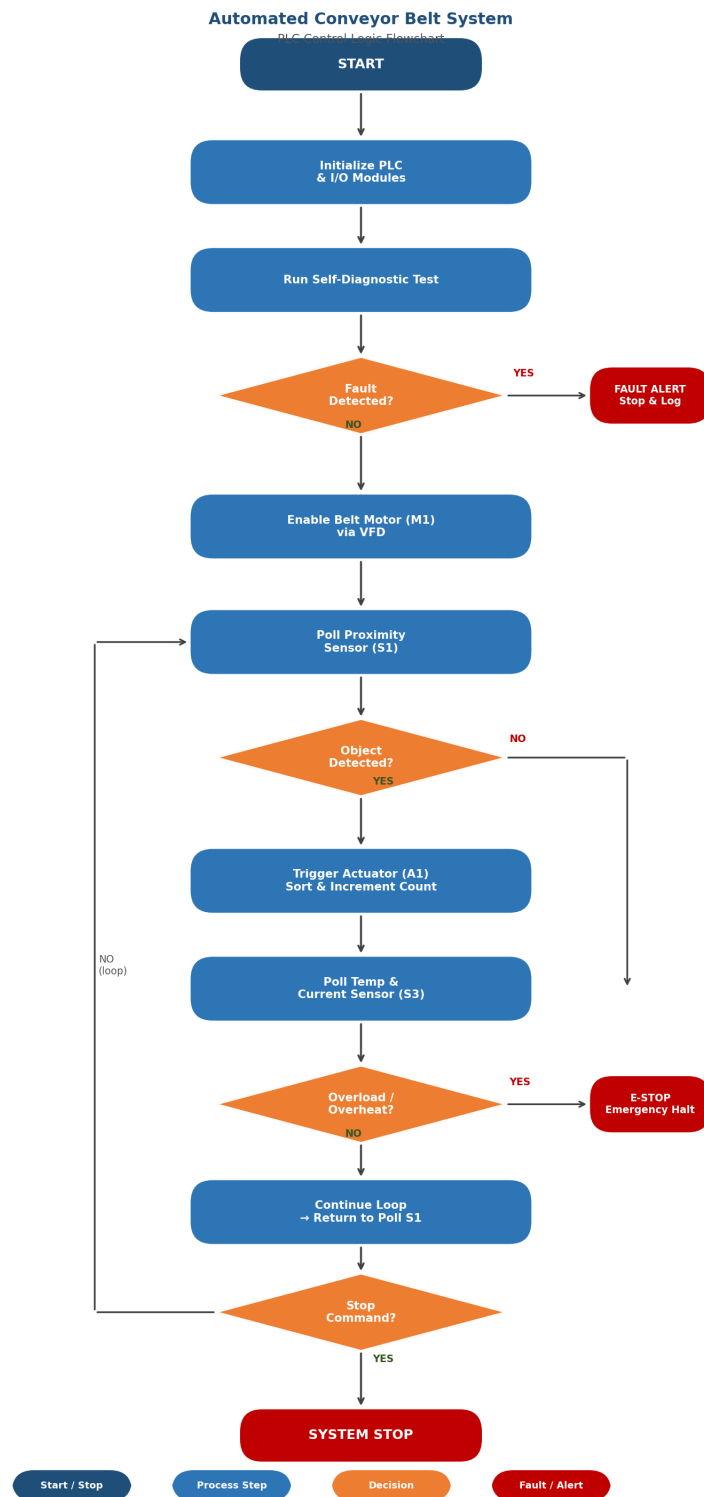
The belt is driven by a three-phase AC motor paired with a Variable Frequency Drive (VFD), allowing smooth speed regulation and energy-efficient starts. Four sensors provide real-time data: a proximity sensor for object detection, a rotary encoder for belt speed, a PT100/CT module for thermal and electrical protection, and a retroreflective photo-eye at the discharge end. Output devices include the motor drive, a pneumatic sorting actuator, and an alarm/HMI indicator tower.

1.1 Key Design Objectives

- Fully automated object detection and sorting with no operator input during normal operation.
- Motor overload and overheat protection through continuous sensor monitoring.
- Fault detection at startup via self-diagnostic routines before belt activation.
- Closed-loop speed regulation via encoder feedback to the VFD.
- Immediate emergency stop (E-STOP) capability at any point during the cycle.

2. Control Logic Flowchart

Below illustrates the complete PLC control logic cycle, from system initialisation through sensor polling, actuator control, fault handling, and the stop condition.



3. Flowchart Step-by-Step Description

Step 1 — System Initialisation

On power-up, the PLC initialises all I/O modules and establishes communication with the VFD and HMI. A self-diagnostic test verifies sensor connectivity and output module status. If any fault is detected at this stage, the system logs the error, triggers the alarm indicator (L1), and holds in a safe stopped state until the fault is cleared by an operator.

Step 2 — Belt Motor Enable

Once diagnostics pass, the PLC energises output Q0.0 to enable the VFD, which starts the belt motor (M1) at the configured set-point speed. The rotary encoder (S2) continuously feeds speed data back to the VFD for closed-loop regulation.

Step 3 — Object Detection and Sorting

The proximity sensor (S1) is polled on every PLC scan cycle. When an object is detected, the PLC increments the production counter and evaluates the sorting criteria. If the object meets the defined parameters, output Q0.1 activates the pneumatic solenoid actuator (A1) to divert the object to the appropriate lane. The event is simultaneously logged to the HMI display.

Step 4 — Protection Monitoring

During each scan cycle, the PLC reads the temperature and current sensor (S3). If motor temperature or drawn current exceeds the configured safety thresholds, the PLC immediately commands an emergency stop: Q0.0 is de-energised, the VFD ramps the motor to a controlled halt, and the alarm (L1) is activated. This prevents motor burnout and mechanical damage.

Step 5 — Loop or Stop

If no stop command has been issued, the control cycle loops back to the sensor polling stage. A stop command can originate from the HMI operator panel, the photo-eye (S4) detecting a backed-up discharge end, or the physical E-STOP button. Upon receiving a valid stop signal, the PLC performs a controlled shutdown of all outputs.

4. Component Specifications

The table below summarises all major hardware components, their types, functions, and PLC I/O pin assignments.

ID	Component	Type	Function	PLC Pin
S1	Proximity Sensor	Inductive/Capacitive	Detects objects on the belt; triggers sort and count routines	I0.0
S2	Speed Encoder	Rotary Encoder	Monitors belt speed; provides feedback to the VFD drive	I0.1
S3	Temp / Current Sensor	PT100 / CT Module	Protects motor against overheating and overload conditions	I0.2
S4	Photo-Eye (End Zone)	Retroreflective	Detects objects at the discharge end; initiates belt stop	I0.3
M1	Belt Drive Motor	3-Phase AC + VFD	Drives the conveyor belt; speed controlled by VFD output	Q0.0
A1	Sorting Actuator	Pneumatic Solenoid	Diverts objects at the sorting station when triggered	Q0.1
L1	Alarm / HMI Indicator	Signal Tower / SCADA	Visual and audible fault notification displayed to operators	Q0.2

5. Safety & Compliance Notes

- All wiring must comply with IEC 60204-1 (Safety of Machinery — Electrical Equipment).
- The E-STOP circuit must be implemented as a hardwired safety relay, independent of the PLC, per ISO 13850.
- Motor overload relays shall be set to 110% of the motor's full-load current rating.
- Lockout/Tagout (LOTO) procedures must be followed during any maintenance on the conveyor or control panel.
- The VFD enclosure must meet a minimum IP54 rating for industrial environments.